

FILE : CH12PP1

```
d = {}
ans = "y"
while ans == "y" or ans == "Y" :
    name = input("Enter employee name: ")
    sal = float(input("Enter employee salary: "))
    d[name] = sal
    ans = input("Do you want to enter more employee names?
(y/n)")
print(d)
```

FILE : CH12PP2

```
str = input("Enter the string: ")
ch = input("Enter the character to count: ");
c = str.count(ch)
print("Count of character",ch,"in",str,"is :", c)
```

FILE : CH12PP3

```
num = int(input("Enter a number: "))
d = {0 : "Zero" , 1 : "One" , 2 : "Two" , 3 : "Three" , 4 :
"Four" , 5 : "Five" , 6 : "Six" , 7 : "Seven" , 8 : "Eight" , 9 :
"Nine"}
digit = 0
str = ""
while num > 0:
    digit = num % 10
    str = d[digit] + " " + str
    num = num // 10
print(str)
```

FILE : CH12PP4

```
d = {}
ans = "y"
while ans == "y" or ans == "Y" :
    name = input("Enter Team name: ")
    w = int(input("Enter number of wins: "))
    l = int(input("Enter number of losses: "))
    d[name] = [w, l]
    ans = input("Do you want to enter more team names? (y/n): ")
team = input("Enter team name for winning percentage: ")
if team not in d:
    print("Team not found", team)
```

```

else:
    wp = d[team][0] / sum(d[team]) * 100
    print("Winning percentage of", team, "is", wp)
w_team = []
for i in d.values():
    w_team.append(i[0])
print("Number of wins of each team", w_team)
w_rec = []
for i in d:
    if d[i][0] > 0:
        w_rec.append(i)
print("Teams having winning records are:", w_rec)

```

FILE : CH12PP5

```

d = {}
ans = "y"
while ans == "y" or ans == "Y" :
    p_name = input("Enter the product name: ")
    p_price = float(input("Enter product price: "))
    d[p_name] = p_price
    ans = input("Do you want to enter more product names? (y/n): ")
ans = "y"
while ans == "y" or ans == "Y" :
    name = input("Enter the product name to search: ")
    print("Price:", d.get(name, "Product not found"))
    ans = input("Do you want to know price of more products? (y/n): ")

```

FILE : CH12PP6

```

days_in_months = {
    "january":31,
    "february":28,
    "march":31,
    "april":30,
    "may":31,
    "june":30,
    "july":31,
    "august":31,
    "september":30,
    "october":31,
    "november":30,
    "december":31
}

```

```

m = input("Enter name of month: ")
if m not in days_in_months:
    print("Please enter the correct month")
else:
    print("There are", days_in_months[m], "days in", m)

print("Months in alphabetical order are:",
sorted(days_in_months))
print("Months with 31 days:", end=" ")
for i in days_in_months:
    if days_in_months[i] == 31:
        print(i, end=" ")
day_month_lst = []
for i in days_in_months:
    day_month_lst.append([days_in_months[i], i])
day_month_lst.sort()

month_day_lst =[]
for i in day_month_lst:
    month_day_lst.append([i[1], i[0]])
sorted_days_in_months = dict(month_day_lst)
print()
print("Months sorted by days:", sorted_days_in_months)

```

FILE : CH12PP6A

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```

days_in_months = {
    "january":31,
    "february":28,
    "march":31,
    "april":30,
    "may":31,
    "june":30,
    "july":31,
    "august":31,
    "september":30,
    "october":31,
    "november":30,
    "december":31
}
m = input("Enter name of month: ")
if m not in days_in_months:
    print("Please enter the correct month")
else:
    print("a) There are", days_in_months[m], "days in", m)

```

```

print("b) Months in alphabetical order are:",
sorted(days_in_months))
print()
print("c) Months with 31 days:", end=" ")
for i in days_in_months:
    if days_in_months[i] == 31:
        print(i, end=" ")
print("-----")
print()
day_month_lst = []
for i in days_in_months: # converting dict to list
    day_month_lst.append([days_in_months[i], i]) # order of dict
changed in list to value:key
print(day_month_lst)
day_month_lst.sort() # dict cannot be sorted
print("---->", day_month_lst)

print()
month_day_lst = []
for i in day_month_lst:
    month_day_lst.append([i[1], i[0]]) # order of list
value:key changed in list to key:value
sorted_days_in_months = dict(month_day_lst) # LIST converted
to dict
print("d) Months sorted by days:", sorted_days_in_months)

```

FILE : CH12PP7

```

n = 10
details = {}
for i in range(n):
    name = input("Enter the name of student: ")
    roll_num = int(input("Enter the roll number of student: "))
    marks = int(input("Enter the marks of student: "))
    grade = input("Enter the grade of student: ")
    details[roll_num] = [name, marks, grade]
    print()
print(details)

```

FILE : CH12PP8

```

x = { "k1" : "v1" , "k2" : "v2", "k3" : "v3"}
inverted_x = {}
for i in x :
    inverted_x[x[i]] = i
print(inverted_x)

```

FILE : CH12PP9

```
d1 = eval(input("Enter first dictionary: "))
d2 = eval(input("Enter second dictionary: "))
print("First dictionary: ", d1)
print("Second dictionary: ", d2)
if len(d1) > len(d2):
    longDict = d1
    shortDict = d2
else:
    longDict = d2
    shortDict = d1
print("overlapping keys in the two dictionaries are:", end=' ')
for i in shortDict:
    if i in longDict:
        print(i, end=' ')
```

FILE : CH12PP10

```
D1 = eval(input("Enter a dictionary D1: "))
print("D1 =", D1)
val = tuple(D1.values())
seen = []
flag = True
for i in val:
    if i not in seen:
        seen.append(i)
        count = val.count(i)
        if count > 1:
            print(count, "keys have same value of", i)
            flag = False
if flag:
    print("No keys have same values")
```

FILE : CH12PP11

```
d1 = eval(input("Enter a dictionary d1: "))
d2 = eval(input("Enter a dictionary d2: "))
print("d1 =", d1)
print("d2 =", d2)
if len(d1) > len(d2):
    longDict = d1
    shortDict = d2
else:
    longDict = d2
    shortDict = d1
```

```
for key in shortDict:
    if key in longDict:
        if longDict[key] != shortDict[key]:
            print("d1 and d2 are different ")
            break
    else:
        print("d1 and d2 are different ")
        break
else:
    print(shortDict, "is contained in", longDict)
```

FILE : CH12PP12

```
D1 = eval(input("Enter a dictionary D1: "))
print("D1 =", D1)
D2 = {}
for key in D1:
    num = sum(D1[key])
    D2[key] = num
print(D2)
```